Indian Institute of Technology Jodhpur
Department of Mathematics

Mathematics II MAL1020

Practice Set 1

AY 25-26 Semester II

---

1. Let $T : P_2 \to P_2$ be defined by

$$T(p(x)) = p(x) - p(1).$$

With respect to the basis $B = \{1, x, x^2\}$:

(a) Find the matrix representation $[T]_B$.
(b) Find a basis for $\ker(T)$.
(c) Find a basis for $\text{Range}(T)$.
(d) Compute the rank and nullity of T.
(e) Verify the Rank–Nullity Theorem.
(f) Show that $P_2 = \ker(T) \oplus \text{Range}(T)$.

2. Let $T : P_3 \to P_2$ be defined by

$$T(p(x)) = p'(x).$$

Consider the bases $B = \{1, x, x^2, x^3\}$ and $C = \{1, x, x^2\}$.

(a) Find the matrix representation $[T]_{C \leftarrow B}$.
(b) Find $\ker(T)$.
(c) Find $\text{Range}(T)$.
(d) Compute the rank and nullity.
(e) Verify the Rank–Nullity Theorem.

3. Let $V = M_{2\times 2}(\mathbb{R})$ and define

$$T(A) = A - A^T.$$

Consider the basis

$$B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}.$$

(a) Find the matrix representation $[T]_B$.
(b) Find a basis for $\ker(T)$.
(c) Find a basis for $\text{Range}(T)$.
(d) Compute the rank and nullity.
(e) Show that $V = \ker(T) \oplus \text{Range}(T)$.

4. Let $T : P_2 \to P_2$ be defined by

$$T(a + bx + cx^2) = a + bx.$$

(a) Find the matrix representation of T with respect to the basis $\{1, x, x^2\}$.

(b) Show that $T^2 = T$.

(c) Find $\ker(T)$ and $\text{Range}(T)$.

(d) Compute the rank and nullity.

(e) Show that $P_2 = \ker(T) \oplus \text{Range}(T)$.

5. Let $T : P_2 \to P_2$ be defined by

$$T(p(x)) = xp'(x).$$

Let $B = \{1, x, x^2\}$ and $B' = \{1, x + 1, x^2\}$.

(a) Find the matrix representation $[T]_B$.

(b) Find the matrix representation $[T]_{B'}$.

(c) Find $\ker(T)$ and $\text{Range}(T)$.

(d) Compute the rank and nullity.

(e) Determine whether $P_2 = \ker(T) \oplus \text{Range}(T)$.

6. Let $T : P_2 \to \mathbb{R}$ be defined by

$$T(p(x)) = p(0) + p(1).$$

(a) Find the matrix representation of T with respect to the basis $\{1, x, x^2\}$.

(b) Find $\ker(T)$.

(c) Compute the rank and nullity.

(d) Find a subspace W such that $P_2 = \ker(T) \oplus W$.

7. Let $V = M_{2\times 2}(\mathbb{R})$ and define

$$T(A) = \frac{1}{2}(A + A^T).$$

(a) Find the matrix representation of T with respect to the standard basis.

(b) Find $\ker(T)$ and $\text{Range}(T)$.

(c) Compute the rank and nullity.

(d) Verify that $V = \ker(T) \oplus \text{Range}(T)$.

8. Let $T : P_3 \to P_3$ be defined by

$$T(p(x)) = p(x) - p(0) - xp'(0).$$

With respect to the basis $\{1, x, x^2, x^3\}$:

(a) Find the matrix representation of T.

(b) Find $\ker(T)$.

(c) Find $\text{Range}(T)$.

(d) Compute the rank and nullity.

(e) Show that $P_3 = \ker(T) \oplus \text{Range}(T)$.